

When the System Becomes Its Own Gravity: On Self-Induced Structural Depletion

Extremum Series Part V — A Fifth Extreme Regime Test in Navigational Cybernetics

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Part V of the Extremum Series.

Previous parts: I — Cannibalism / II — Suicide / III — The Anti-Extreme / IV — Structural Implosion

These phenomena are not the subject of this work. They are used only as extreme regime tests revealing the architecture of identity.

Prologue: The Missing Actor

Part IV described structural implosion — a regime in which the admissible interior contracts around a system that never violated anything. No event, no actor, no crisis. The pressure was environmental and diffuse: weather, market friction, institutional weight, the cost of existing under load. The system depleted silently because load was sustained and τ is monotone.

But implosion left a question unanswered. In implosion, the pressure comes from outside — or more precisely, from nowhere in particular. The environment loads the system. The system absorbs it. The asymmetry is between the system and its context, but the context has no agency, no policy, no self-reinforcing logic.

What happens when the source of pressure is not the environment but the system itself? When the policy that generates structural burden and the structure that absorbs it belong to the same entity? When the system builds its own gravity — and then lives under it?

1. The Fifth Mode

There is a class of structural depletion that is neither externally imposed (implosion) nor instantaneously self-inflicted (suicide). It is a sustained, self-generated regime in which the system's own policy progressively narrows its admissible interior — not through a single decision, but through a persistent behavioral pattern that accumulates structural burden over time.

The system does not break. It does not choose to die. It builds the conditions of its own exhaustion incrementally, through a policy that it maintains, reinforces, and often defends.

I am going to call this **self-induced structural depletion**.

This is not suicide. Suicide (Part II) is a puncture — one act, one crossing, one moment in which the system authorizes a transition that destroys its admissible space. Self-induced depletion is not a moment. It is a regime. The system does not cross a boundary. It constructs the boundary closer to itself, day after day, through its own continued operation.

And this mode has two structurally distinct sub-regimes that share the mechanism but differ in everything else.

2. Sub-Regime A: The Unnecessary Noose

In the first sub-regime, the system constructs the conditions of its own depletion without structural necessity. The policy that narrows the admissible interior is not required for survival. It is adopted, maintained, and defended by the system as if it were load-bearing — but it is not.

Anorexia is the clearest example. The system restricts its own input — food, energy, metabolic substrate. Each restriction is a policy decision. Each narrowing is incremental. Each is locally coherent: the system has reasons, frameworks, internal logic that justify every step. The metrics it watches — weight, control, shape — remain "on track". The metrics it does not watch — bone density, organ function, cognitive capacity, structural reserve — degrade silently.

The policy does not pay a structural cost for continuing. It is a rule. Rules do not deplete. The body depletes. The asymmetry is internal: the policy is the system's own inexhaustible gravity.

This pattern repeats across scales:

A perfectionist raises the standard after every success. Each new threshold is a self-imposed policy. The bar never falls. The capacity beneath it does. Eventually nothing passes the system's own filter. The admissible interior contracts to the empty set — not because nothing is good enough in any objective sense, but because the filter is the system's own construction and the system cannot stop tightening it.

An organization optimizes for efficiency — cuts R&D, removes redundancy, compresses response time. Each cut is locally rational. Each improves the metric it was designed to improve. Each removes a piece of the structural capacity that will be needed when the metric stops being the right question. Boeing built the textbook: a decade of cost-cutting with green KPIs, until the aircraft started falling.

An authoritarian state tightens control for "stability" — centralizes decisions, silences feedback, expels dissent. Each step increases measured stability. Each narrows the space of admissible system responses. The late Soviet Union suffocated itself with its own restrictions, each one imposed to prevent collapse, each one making the collapse it was designed to prevent more certain.

In all cases: the policy is the system's own. Nobody is doing this to it. The policy does not tire. The structure does.

3. Sub-Regime B: The Necessary Wound

The second sub-regime is structurally more interesting and architecturally more important.

Here, the system also builds the conditions of its own depletion. But it does so because **the alternative is immediate collapse**. The self-inflicted structural burden is not unnecessary. It is the only admissible continuation available.

A surgeon amputates a gangrenous leg. This is self-inflicted structural damage: the system permanently reduces its own capacity. Φ jumps upward. The admissible interior contracts. But without the amputation, $\tau \rightarrow 0$ within days. With it, the system survives — diminished, permanently burdened, but with remaining budget to spend.

Chemotherapy poisons the entire organism. Every cell pays a structural cost. Φ accelerates. The body degrades visibly, measurably, painfully. But the alternative is a tumor that will exhaust τ faster than the

poison does. The system chooses the slower poison over the faster one. It accelerates its own structural burden to extend the time before the budget runs out.

A company in crisis cuts entire divisions — not because it wants to, but because runway is measured in weeks. Structural capacity drops permanently. But without the cut, there is no next quarter. The system trades structural depth for temporal extension.

An army cedes territory — not as strategy, but because holding it costs more τ per day than retreating does. It burns its own structural depth to preserve the core. Retreat is self-inflicted contraction of the admissible interior. But every alternative contracts it faster.

A person in winter burns furniture to stay warm. Each piece burned is structural capacity that will not return. The house becomes less of a house. But without heat, there is no person to inhabit it.

In every case: the system **accelerates Φ deliberately** in order to **keep τ above $\tau_{\min}$ for longer**. This is the structural signature of the necessary wound.

4. The Formal Structure

What makes Sub-Regime B architecturally distinct — and what makes it a genuine contribution to the extremum series — is its relationship to admissibility.

In all previous modes, the system either exits its admissible space (cannibalism, suicide), renegotiates it (anti-extreme), or watches it contract around it (implosion). In Sub-Regime B, the system **deliberately damages its own structural capacity because the admissibility predicate leaves no other option**.

Formally:

Let $A(t)$ denote the set of admissible continuations at time t . Let $U_{\blacksquare}$ denote the set of actions that do not increase Φ beyond its natural accumulation rate. Let U_d denote the set of actions that impose additional self-inflicted structural burden ($\Delta\Phi_{\text{self}} > 0$).

In Sub-Regime B, the following condition holds:

$$A(t) \cap U_{\blacksquare} = \emptyset$$

There are no admissible continuations that do not involve self-damage. Every trajectory in $U_{\blacksquare}$ leads to $\tau(t+1) < \tau_{\min}$. The only trajectories that keep τ above $\tau_{\min}$ are those in U_d — trajectories where the system pays an additional structural cost now to avoid a larger structural cost immediately.

This is not optimization. The system is not minimizing a loss function. There is no gradient signal pointing toward the "best" self-inflicted wound. The admissibility predicate simply excludes all non-self-damaging trajectories. What remains is the set of wounds the system can survive.

This is admissibility in its purest and most brutal form: the gate does not recommend. It does not rank. It does not suggest. It says: "these trajectories are admissible, those are not". And when only the self-damaging trajectories pass the gate, the system either damages itself or ceases to exist.

The non-causality is precise: the admissibility predicate does not cause the self-damage. It constrains the realization space such that only self-damaging trajectories remain. The system's own survival imperative — the drive to keep $\tau > \tau_{\min}$ — does the rest. Admissibility provides no gradient, no direction, no guidance. It provides a boundary. The system walks into the wound because every other direction has already been walled off.

5. Why This Is Not Suicide

Suicide (Part II) is a single act that destroys the admissible space. The system authorizes its own termination.

Self-induced depletion in Sub-Regime B is the opposite. The system damages itself **in order to continue existing**. It does not authorize termination. It authorizes degradation as the price of persistence. Every self-inflicted wound is an act of survival, not surrender.

This produces a paradox that is not logical but structural: the system that survives longest may be the one that has inflicted the most damage on itself. The system with the highest accumulated self-inflicted burden Φ_{self} may have the longest trajectory — because each unit of self-inflicted burden purchased time that would otherwise not have existed.

The system cannot outlast its own structural budget — but it can trade pieces of itself for more time under load. This is not heroism. It is not pathology. It is the arithmetic of bounded existence, performed by the system on itself.

6. Why This Is Not Anti-Extreme

The anti-extreme (Part III) is a voluntary override of identity under three conditions: structural viability, multiple admissible continuations, and a deliberate choice not to take the identity-preserving path.

Sub-Regime B violates the second condition. There are not multiple admissible continuations. There is one class: self-damage. The system does not choose to override its identity. It is forced to consume itself because the admissibility gate has already closed every other door.

The anti-extreme is freedom exercised at the edge. Sub-Regime B is necessity accepted at the edge. They look similar from outside. Structurally, they are opposites.

7. The Structural Pair

This text and Part VI form a pair, just as Parts I and II did.

Cannibalism and suicide were the first pair: external boundary dissolution versus internal boundary collapse. Same architecture of identity, tested from opposite directions.

Self-induced depletion and the mode that follows it — asymmetric temporal exhaustion — are the second pair: **asymmetric exhaustion by an internal inexhaustible policy** versus **asymmetric exhaustion by an external inexhaustible source**. In both, the source does not deplete. Here, the source is the system's own operational logic. In Part VI, the source will be outside.

And Sub-Regime B adds a layer that has no analogue in the series: the system that damages itself to survive is the system that has understood, structurally, that the cost of continuation exceeds the cost of self-inflicted harm. It has internalized the terminal admissibility horizon and chosen to trade structure for time.

This is not heroism. It is not pathology. It is the arithmetic of bounded existence under inexhaustible load, performed by the system on itself.

8. The Shape of the Series

Part I was about what happens when you consume the other.

Part II was about what happens when you consume yourself.

Part III was about the conditions under which self-consumption can be structurally authorized.

Part IV was about what happens when there is nothing left to consume — and the system doesn't know yet.

Part V is about what happens when you are the thing that does not run out — and about when becoming your own gravity is the only way to stay alive.

This work is part of Navigational Cybernetics 2.5 (NC2.5), a formal theory of long-horizon adaptive systems.

Extremum Series: I — Cannibalism / II — Suicide / III — The Anti-Extreme / IV — Structural Implosion / V — Self-Induced Structural Depletion

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